

Portfolio Flood Risk Assessment: Key Components

Technical Documentation Team

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1 Core Model Components

1.1 Property Valuation Model

- PropertyValuationModel class
 - Feature preprocessing pipeline
 - XGBoost regression model
 - PCA dimensionality reduction
- Spatial analysis components
- Market factor calculators

1.2 Flood Risk Model

- FloodRiskModel base class
- EnhancedFloodRiskModel extension
- Spatial correlation engine
- Impact calculation system

2 Supporting Components

2.1 Data Management

- ProjectPaths utility
- GeoDataFrame handlers
- Data validation systems

2.2 Analysis Tools

- Spatial clustering engine
- Risk concentration calculator
- Stress testing framework

2.3 Visualization Components

- Interactive mapping system
- Risk heatmap generator
- Correlation visualizer

3 Integration Interfaces

- Portfolio data standardizer
- Risk metric aggregator
- Report generation system